Exercises 7.8 5, 7, 9, 11, 13, 15, 17, 19, 23, 25, 31, 33, 37, 39, 49, 51, 53, 55, 59.

5. $\int_1^\infty 2x^{-3}\,dx$

$= \lim_{t\to\infty} \int_1^t 2x^{-3}\,dx$

$= 2\lim_{t\to\infty} \left[\frac{x^{-2}}{-2}\right]_1^t$

$= -\lim_{t\to\infty} \left[x^{-2}\right]_1^t$

$= 1$

9. $\int_{-2}^\infty \frac{1}{x+4}\,dx$

$= \lim_{t\to\infty} \int_{-2}^t \frac{1}{x+4}\,dx$

$= \lim_{t\to\infty} \left[\ln|x+4|\right]_{-2}^t$

$= \infty$

... is divergent

19. $\int_{-\infty}^{\infty} xe^{-x^2}\,dx$ 两端都发散

$$= \int_{-\infty}^{0} x e^{-x^2}\,dx + \int_{0}^{\infty} x e^{-x^2}\,dx$$

$$= \lim_{t\to-\infty} \int_{t}^{0} x e^{-x^2}\,dx + \lim_{s\to\infty} \int_{0}^{s} x e^{-x^2}\,dx$$

let $u = -x^2$; $\frac{du}{dx} = -2x \Rightarrow dx = \frac{du}{-2x}$

when $x = 0 \quad u = 0$

$x = -\infty \quad u = -\infty$

$x = 0,\ u = 0$

$x = \infty,\ u = -\infty$

$$= \lim_{t\to-\infty} \int_{-\infty}^{0} e^u \cdot x \cdot \frac{du}{-2x} + \lim_{s\to\infty} \int_{0}^{s} e^u \cdot x \cdot \frac{du}{-2x}$$

$$= \lim_{t\to-\infty} -\frac{1}{2}\left[e^u\right]_{-\infty}^{0} - \frac{1}{2}\lim_{s\to\infty}\left[e^u\right]_{0}^{s}$$

$$= -\frac{1}{2} + \frac{1}{2} \qquad -\frac{1}{2}(0-1)$$

$= 0$ can cancel $-\frac{1}{2}(1-0)$

31. $\int_{-\infty}^{0} \frac{z}{z^4+4}\,dz$

$\frac{d}{dx}\tan^{-1}x = \frac{1}{1+x^2}$

let $u = z^2$; $\frac{du}{dz} = 2z \Rightarrow dz = \frac{du}{2z}$ $\quad \frac{1}{a^2+x^2} \Rightarrow \frac{1}{a}\tan^{-1}\left(\frac{x}{a}\right)$

$z = 0$, $u = 0$

$z = -\infty$, $u = +\infty$

$$= \lim_{t\to+\infty} \int_0^t \frac{z}{u^2+4} \cdot \frac{du}{2z}$$

$$= \frac{1}{2} \lim_{t\to+\infty} \int_0^t \frac{1}{u^2+4}\,du$$

$$= \frac{1}{2}\times\frac{1}{2} \lim_{t\to\infty} \left[\tan^{-1}\left(\frac{u}{2}\right)\right]_0^t$$

$$= \frac{1}{4}\left(-\frac{\pi}{2}\right) = -\frac{\pi}{8}$$

33. $\int_0^\infty e^{-\sqrt{y}}\,dy$

let $u = -\sqrt{y}$; $\frac{du}{dy} = -\frac{1}{2\sqrt{y}}$ $\Rightarrow dy = -2\sqrt{y}\,du$

$y = -u^2$ $= -2u\,du$

$y = \infty$, $u = -\infty$

$y = 0$, $u = 0$

$= \lim_{t \to -\infty} \int_t^0 e^u \cdot (-2u)\,du$

$= -2 \lim_{t \to -\infty} \int_t^0 u \cdot e^u\,du$

let $a = u$, $dv = e^u du \Rightarrow v = e^u$

$da = du$

$= -2 \lim_{t \to -\infty} \{ [u \cdot e^u]_t^0 - \int_t^0 e^u\,du \}$

$= -2\,(0 - 1)$ $-(1-0)$

$= 2$ e^u

39. $\int_{-2}^{3} \frac{1}{x^4}\,dx$

$= \int_{-2}^{0} \frac{1}{x^4}\,dx + \int_{0}^{3} \frac{1}{x^4}\,dx$

$\int_{-2}^{0} \frac{1}{x^4}\,dx = \lim_{t \to 0^-} \int_{-2}^{t} \frac{1}{x^4}\,dx$

$= \lim_{t \to 0^-} \left[-\frac{x^{-3}}{3} \right]_{-2}^{t} = \lim_{t \to 0^-} \left[\right.$

$= \infty$ Divergent